

1. Coconut Climber

Special Features:

- Useful for climbing coconut trees for harvesting nuts, cleaning and other operations.
- Any unskilled person including ladies can climb the coconut trees using this unit.
- Requires 1.5 minutes to climb a tree of 30 to 40 ft height.

Cost of the Unit: Rs.2,000/-

Capacity: 50 to 60 trees/day

Cost of Operation: Rs.1.50/tree

Manufacturer: TNAU, Coimbatore

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2. Coconut Dehusker

Kerala Agricultural University has developed a very simple and effective tool for easy dehusking of coconuts. It takes about 8 to 12 seconds for dehusking a nut depending on the type and maturity. The tool is light, very simple to use and handy and can be used both indoors and outdoors.

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3. Coconut Dehusking Machine

CoCoMaN coconut dehusking machine model COM11 peels off the coconut husk from coconut fruit to obtain dehusked coconut fruit via hydraulically controlled dehusking devices. An operator is required to handle the machine during the dehusking process.

Manufactured by: Method Machine Works, a Malaysia-based private limited. Company is one of the best known manufacturer and exporter of industrial machineries and equipments.

Address: Suite, 26.3, Menara IMC, No. 8, Jalan Sultan Ismail, Kuala Lumpur, WP Kuala Lumpur 50250 Malaysia. www.indiamart.com/methodmachine/products.html

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4. Coconut Deshelling Machine

This machine detaches the hard coconut shell from dehusked coconut fruit via a rotating disc cutter. An operator is required to hold the coconut fruit during the deshelling process. It comes with 1 or 2 deshelling stations.

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5. Coconut Paring Machine

This machine scalps off the soft and brownish skin, or testa, of deshelled coconut flesh. An operator is required to feed the pieces of deshelled coconut flesh into the machine during the paring process.

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6. Coconut Grinding Machine

This machine grinds deshelled coconut flesh into fine desiccated coconut. An operator is required to feed the pieces of deshelled coconut flesh into the machine during the grinding process. It can also be used to grind copra (dried coconut) to fine ground copra flakes.

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7. Coconut Milk Extracting Machine

This machine automatically squeezes the fine desiccated coconut via a rotating screw to obtain coconut milk.

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8. Copra Oil Press

This range of machines are used to compress the fine ground copra flakes to obtain copra oil. Some models have pressing & filtering mechanisms intergrated & also equipped with heaters to improve the output efficiency. It can also be used for oil extraction for peanut, beans, rape seeds, cotton seeds, sesame seeds, olive, sunflower seeds etc.

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9. Agricultural Waste Fired Mechanical Dryer

Function: To burn agricultural wastes and to produce hot air.

Specification:

(a) Overall Dimension: 3100 x 890 x 200mm

(b) Capacity: 1 tonne/day

(c) Power Required: 2 hp Electric Motor

General Information: This unit consists of an agricultural waste fuelled furnace, blower and rectangular metal bin for keeping the grains. The fire tube type furnace can generate heat by burning any agricultural waste. The unit can produce hot air upto 80°C by suitably mixing the hot air with ambient air through an adjustable butterfly valve. Through a flexible duct, the hot air is supplied to the drying bin.

Cost of the Unit: Rs.1,00,000/-

Cost of operation: Rs.20/h

Salient Features:

- All types of agricultural wastes and crop residues can be used as fuel.
- Suitable for drying pods, grains, coconut kernel, etc.
- Air temperature and flow rate can be adjusted and controlled.

Manufacturer: TNAU, Coimbatore

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10. Power Tiller Operated Heavy Duty Auger Digger

Function: To dig holes for planting tree saplings

Specification:

i. Type: Screw auger type

ii. Power requirement: 8 to 10 hp power tiller

iii. Overall dimensions: 400 x 635 x 1635 mm

iv. Weight: 50 kg

v. Capacity: 25 -30 holes per hour

General Information: It consists of a spiral auger of 225 mm dia and 100 mm pitch actuated by a rack and pinion arrangement. The drive for the circular motion of the auger is effected through belt pulley and bevel gear transmission with a ratio of 1:1 from the engine pulley of the power tiller directly. The entire assembly is mounted on a rectangular frame with necessary bearings and fixtures. The hand wheel provided at the side of the unit can be effectively used for the depth control. For increased size of holes, replaceable type larger diameter auger bits of 250, 275 and 300 mm can be used. A balancing frame attached to the hitch bracket assembly in the rear and a support wheel fitted in the front portion of the unit counter the down suction offered by the auger while digging, ensure easy movement in the field, reduction of drudgery and easy manoeuvrability.

Cost of the Unit: Rs. 20,000/-

Salient features:

- Results in 16.0 and 91.0 per cent saving in cost and time when compared to manual digging of holes
- It can reach spots where tractor entry and manoeuvrability are difficult

Manufacturer: TNAU, Coimbatore

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11. Briquetting Machine

Function: To make briquettes from coir pith.

Specification:

(a) Overall dimension: 765 x 350 x 990 mm

(b) Capacity: 125 kg/h

(c) Power required: 5 hp Electric Motor

General Information: The machine consists of a feed hopper, an over hanging screw shaft, a barrel housing and an extruder die pipe. The screw shaft is mounted at one end by ball bearings and the other end is over hanging freely. The barrel assembly has been housed over the screw shaft and mounted on a stand. The free end of the screw shaft is tapered to a length of 30 cm and housed inside an extruder pipe of same length. The screw shaft is coupled directly to a reduction gear unit (10:1) and is driven by a 5 hp electric motor. Fibre free coir pith and cowdung are mixed at 6:1 ratio with required quantity of water so that it will form briquettes during operation.

Cost of the Unit: Rs.30,000/-

Cost of operation: Rs.20/h

Salient features:

- Continuous horizontal type extrusion machine.
- Briquettes are formed by compression.
- Produces hollow briquettes with a calorific value of 3000 K.cal/kg.

Manufacturer: TNAU, Coimbatore

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12. Coir Pith Applicator

Function: For subsoil mulching.

Specification:

i. Type: Tractor operated, built around the chisel plough

ii. Power requirement: 35 to 45 hp tractor

iii. Overall dimensions: 840 x 1000 x 12390 mm

iv. Weight: 50 kg

v. Capacity: 0.5 ha per day

General Information: The subsoil coir pith mulching machine was built around chisel plough which formed the tool for loosening the soil and also provided a frame for mounting the attachment. Coir pith was fed from a hopper through a rotary vane type feeding device and was funneled into the furrow bottom. A pair of furrow opener wings hold the furrow opener behind the chisel plough for placement of the subsoil mulch. The rotary vane type metering device was manually operated and ensured uniform placement of coir pith inside the furrow. The unit was found to have a draft of 500 kg under red soil conditions and 600 kg under black soil conditions.

Cost of the Unit: Rs. 8,500/-

Salient Features: The uniformity of feed rate was found to be at 90%

Manufacturer: TNAU, Coimbatore

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13. Compost Pelletizer

Function: To make pellets from compost.

Specification:

(a) Overall dimension: 1000 x 900 x 900 mm

(b) Capacity: 100 kg/h

(c) Power required: 5 hp Electric Motor

General Information: The compost mixture is fed into the barrel housing the screw auger. The screw conveys the compost against the die assembly and extrude it as pellets. A rotating knife cuts the pellets into small pieces. The pellets are passed over an oscillating tray and collected at the bottom as globules.

Cost of the Unit: Rs.25,000/-

Cost of Operation: Rs.15/h

Salient Features:

- Pellets of uniform size can be obtained.
- Pellets can be easily distributed in the field as manure.
- More suitable for soil application at specific locations.

Manufacturer: TNAU, Coimbatore

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14. Sprayer For Coconut Trees

Function: To spray chemicals in coconut trees and orchards

Specification:

i. Type: Mounted implement

ii. Power requirement: 35 hp tractor

iii. Overall dimensions: 65 x 65 x 1950 mm

iv. Weight: 200 kg

v. Capacity: 35 trees per hour

General Information: The unit has a telescopic (62 and 37 mm) G.I. pipes which can extend from 9 m to 14 m height by winding a cable. The telescopic G.I. pipes are mounted on a 'L' angle frame work hitched to the tractor through the three point linkage connection. At the base of the frame, a 200 lit capacity chemical tank is mounted. The commercially available

ASPEE triplex reciprocating pump is mounted on a base plate near the PTO shaft of the tractor. At the top of the 37 mm G.I. pipe, two spray guns with nozzles for cone or jet spray are hinged so that they can be moved up and down by ropes from the ground level. In addition the spray guns can be rotated about the vertical axis of the telescopic pipe arrangement. These features of the sprayer ensure that the entire canopy area of the tree is easily sprayed.

Cost of the Unit: Rs. 45,000/-

Salient Features:

- The total height of spray is 55 feet.
- For easy transport, the unit can be kept in a horizontal position by folding the frame with telescopic pipes and the same can be erected vertically by winding a cable.
- The sprayer can be used in orchard trees and for spraying in field crops, by bifurcating the delivery section into spray lines with spray lances and spray guns.

Manufacturer: TNAU, Coimbatore

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15. Production of Particle Board From Coirpith

Function: To utilize coir pith for the production of particle board

General Information: Coir pith being a waste disposed from the coir industries, pose the danger of pollution and health hazard. Being a lingo-cellulosic material, coir pith can be utilized for the production of particle board. The process of production of coir pith particle board involves drying of the coir pith and resination using phenol formaldehyde (16%) or urea formaldehyde (20%). The resinated pith is formed into a mat of desired thickness and hot pressed at 120°C for 15-20 minutes. This board after the formation is trimmed along the sides and suitable for use as table top and providing partition and other light works. The laminated boards will be suitable for exterior use, as the water absorption and swelling character are more for the coir pith boards.

Cost of Operation: Rs.40/sq.ft. of the board.

Salient Features:

- Finds utilisation for the coir pith
- The boards are suitable for interior use as table top and for providing partition.

Manufacturer: TNAU, Coimbatore

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16. TNAU Power Weeder with Rotary and Tined Attachments

A power weeder was developed at TNAU, Coimbatore. The equipment consists of a 5.4 hp, 34 kg light weight diesel engine mounted on the frame. From the engine, the power is transmitted to the transmission gear box and then to the ground wheels and rotary weeder. The machine is suitable for wide spaced crops such as cotton, tapioca, sugarcane, grapes, coconut, arecanut and other orchard and plantation crops. The row mounted hoe and sweep type weeders have three staggered tynes with provision for adjusting row-to-row spacing. The engine is mounted in front of the drive wheels for balancing of the machine. The cost of the machine is Rs 50,000/- and its cost of operation is Rs 380/ha. The effective field capacity and field efficiency were 0.09-0.11 ha/h and 85%, respectively.

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17. Coconut Snowball Tender Coconut Machine: CPCRI

The latter has also developed a coconut bunch support to be fixed on the trunk of the coconut tree to prevent the buckling of coconut bunches, and a coconut dehusking machine operated manually for reducing the drudgery and fatigue of traditional dehusking operation.

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18. Power Operated Coconut Dehusker

Features: The machine is used for dehusking of coconut. It consists of semi-circular stationary concave and rotating drum with knives. The electric motor (3 hp) with 10:1 speed reduction unit is used to drive the drum. The power is transmitted from motor to reduction unit using V -belt and further power from reduction unit to rotating drum is transmitted with chain and sprockets. During operation, the small knives bite the coconut fed through the inlet. At the outlet the nut is fed through two rollers to remove parts of husk remaining on the nut after dehusking.

Specifications:

Stationary concave width (rom)	250
Rotating drum width (rom)	250
No of rollers	2
Size of roller (1 x diameter) (rom)	280 x 955
Number of teeth on rollers	15
Frame (1 x b x h) (rom)	1020 x 380 x 970
Length of knives (rom)	12
Power source (hp)	3, Electric motor

Performance Results:

Rotary drum speed (rpm)	22-32
Time taken for dehusking one coconut (s)	10-20
Moisture content (%)	25.40 - 60.2
Torque (kg-m)	67.14 - 96.35

Uses: It is used for dehusking of coconut.

Source: <http://agricoop.nic.in>

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19. Manually Operated Coconut Splitting Machine

Features: The manually operated device- is used for splitting dehusked coconuts. It consists of pedestal, pivoted long handle, cutting knife and a platform. Lifting the handle raises the knife. The dehusked nut is placed on the platform and then the knife is lowered to cut the nut. The water of coconut is collected after splitting the nut.

Specifications:

Overall dimensions (lxbxh) (mm): 650 x 650 x 1450

Pivoted handle length (mm): 550

Performance Results:

Force required to cut the nut (kg): 155

Effort at the handle (kg): 30

Mechanical advantage: 5.17

Uses: It is used for splitting dehusked coconut kernel.

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20. Cocotap

COCOTAP is a patented quality product, made to last from food grade stainless steel 1.6mm tube, professionally milled and hand finished in Australia. It is easily re-sharpened and cleaned and can be hygienically used in premises serving the public. The T-bar handle is hinged so that the COCOTAP folds to a straight tube for easy use and storage.

Features: Sharpened 1/2" diameter shaft with clearing port and 5/8" diameter handle. The total length is 170mm and makes a hole large enough for 2 straws. It is perfect for the average size hand and adding cocktail mixes. There is a short braided lanyard for hanging, and a protective cap with cleaning tool attached for convenience.

How to use the COCOTAP:

1. Clutch the handle so that the solid side rests in the base of your palm. Use your knuckles, as in the picture, or thumb and forefinger to stabilize the shaft while pushing and twisting the spike into the middle of the coconut. Aim perpendicular to the inner shell so that the force applied pierces, rather than skips off, the inner shell. Remove by twisting out and drink through a straw.
2. If the coconut inner shell is too well developed to twist and push straight through, stop forcing and hold the COCOTAP handle upright. Then lightly strike the end, using a hammering device. Alternatively, invert the coconut with COCOTAP aimed at core then tap the end of the handle on a solid surface using the weight of the coconut as the "hammer" onto the COCOTAP spike. *Note: The COCOTAP is not designed to penetrate fully matured coconuts, except through the soft eye once dehusked* Remove the COCOTAP by the opened handle, twisting it out, then insert a drinking straw or two. If a straw is not readily available, simply tap two holes and drink or pour directly from the lower hole.
3. If necessary, clean the core of husk fibre from the COCOTAP through the inspection port, with the tool supplied, or this can be done using a coconut stalk of suitable diameter or any slightly flexible rod, screw driver etc. *Note: depending on technique, coconut maturity, shell and husk density, the core often does not lodge in the COCOTAP.*